

Project Design Phase-I
Proposed Solution Template

Parameter	Details
Date	07 March 2026
Team ID	
Project Name	VoltEdge: EV Analytics Portal
Maximum Marks	2 Marks

Proposed Solution Details

S.No.	Parameter	Description
1.	Problem Statement	As of 2026, stakeholders in the Indian EV ecosystem—urban planners, fleet managers, and consumers—still lack a unified platform to analyze disparate data. This leads to inefficient charging infrastructure placement, unpredictable battery maintenance costs, and difficulty in comparing EV models based on real-world efficiency rather than manufacturer claims.
2.	Idea / Solution Description	VoltEdge is an integrated analytics portal developed as a rapid-response solution in March 2026. It utilizes a PostgreSQL database for structured data storage and Tableau for high-fidelity interactive visualizations. The system is served via a Flask web application, offering dashboards for regional charging density mapping and brand-wise performance comparisons.
3.	Novelty / Uniqueness	VoltEdge features a live Mechatronic-Data-Web bridge. It specifically uses the Tableau JavaScript API to embed interactive "Data Stories" directly into a custom-branded portal. It is uniquely designed for three key personas: the City Planner (Aarav), the Fleet Manager (Meera), and the Consumer, providing tailored insights from a single source of truth.
4.	Social Impact	The solution promotes the adoption of green energy by reducing "range anxiety" through transparent infrastructure mapping. By helping cities optimize power grids for EV loads, it directly supports national sustainability goals and improves public confidence in eco-friendly transitions.
5.	Business Model	SaaS & B2G Licensing: VoltEdge generates revenue through a subscription model for logistics/taxi fleet operators (for efficiency monitoring) and data licensing to municipal governments for urban infrastructure planning.
6.	Scalability	The PostgreSQL backend ensures the system can scale from local city-level data to national-level fleet management. The modular Flask architecture allows for the future addition of AI/ML modules for predictive battery analysis and IoT sensor integration.

Technical Breakdown for the Solution:

- Data Core: PostgreSQL manages the relational integrity of vehicle specifications and charging logs.
- Logic Layer: Tableau performs complex aggregations (e.g., Calculated Fields for "Efficiency by Brand").
- Presentation Layer: The VoltEdge Web Portal (Flask-based index.html) provides a responsive environment for stakeholders to interact with the data.

Team Member Roles (March 2026 Sprint):

- Lavani Sardar: Developed the core PostgreSQL schema and ensured the technical feasibility of the data-to-dashboard pipeline.
- Sanskruti Roman (Lead): Defined the "Social Impact" and "Novelty" while architecting the Flask frontend and embedding the visualization portal.
- Sanskruti Sawant & Sayali Firake: Conducted market research to define the "Problem Statement" and formulated the revenue-generating "Business Model."

3-Day Sprint Milestone (For your reference):

- Day 1 (March 6): Data collection, PostgreSQL database setup, and SQL query optimization.
- Day 2 (March 7): Tableau Dashboard design, creating 11 visualizations, and publishing to Tableau Public.
- Day 3 (March 7): Flask backend development, UI styling (CSS), and final report documentation.